

I • Master Field Equation

Stationary: $(\nabla^2 + k^2)G(x, x') = -\delta^3(x - x')$

Time-dep.: $(\frac{1}{v_s^2}\partial_t^2 - \nabla^2 + k^2)\Phi_{\mu\nu} = -J_{\mu\nu}$

Retarded propagator ($\tau > 0$): $G_R(r, \tau) = \frac{v_s}{4\pi r} e^{-kv_s\tau} \delta(\tau - r/v_s) \theta(\tau)$

Memory kernel: $K(\tau) = K_0 e^{-\tau/\tau_m} \theta(\tau)$, $\tau_m = 1/kv_s$

General solution: $\Phi(x, t) = \Phi^{(0)} + \int_{-\infty}^t G_R \cdot J$

II • Type-Theoretic Architecture

Σ -type (fiber bundle): $\text{SomaField} \equiv \sum_{\sigma: \text{Scale}_{20}} \text{Sub}(\sigma)$

Zoom Operator: $\Lambda : (\sigma : \text{Scale}_{20}) \rightarrow \text{Sub}(\sigma) \rightarrow \text{Sub}(\sigma+1)$

Causality as proof: $G_R : (t \ t' : \text{Time}) \rightarrow (t' < t) \rightarrow \text{Sp} \rightarrow \text{Sp} \rightarrow \text{Field}$

BFSS cortex: $\text{CortexCoords} \equiv \text{eig}(X)$, $X^\dagger = X$, $X \in M_{3 \times 3}(\mathbb{C})$

III • Attractor Dynamics

Hopfield: $H(\mathbf{e}) = -\frac{1}{2} \mathbf{e}^\top W \mathbf{e}$

FM-HN (*Lean 4*): $\beta(t) = \beta_0 + \kappa |\Phi|^2 \xrightarrow{\Phi \rightarrow 0} \beta_0$

Consciousness: spectral gap opens at $|\phi| > T_c$

Kramers MFP: $\langle T \rangle = \frac{2\pi}{\omega_A \omega_B} e^{\Delta V/D}$

Trauma asymmetry: formation $\ll$ dissolution (above vs below barrier)

WKB: $P \approx \exp(-\frac{2}{\hbar_{\text{geo}}} \int \sqrt{V - E} dq)$

Rapport (Huygens): $|\omega_{\text{ext}} - \omega_0| < \Delta\omega(\kappa)$

SHO (*physlib*): $\omega^2 = k/m$, $\omega^2 = v^2 k^2$

IV • 11D Decomposition

$M_{11} = M_4 \times X_7$, $X_7 = D_{5-7} \times D_8 \times D_{9-11}$

- D_{1-4} Spacetime · D_{5-7} Propagator/D3-brane
- D_8 Limbic · HW orbifold [-1 =body, $+1$ =mind]
- D_{9-11} Cortex/BFSS eigenvalues
- **Cosmo:** $\kappa_{\text{bio}} \rightarrow 0 \Rightarrow \text{GR}$; $\Lambda \equiv \langle \text{tr } \Phi \rangle_0$

V • Key Identifications

- **SHO** $\equiv$ **Green's fn** — string = impulse response
- **Nash** $\equiv$ **Hopfield min** — $\arg \min H$
- **Rights** $\equiv$ **topo. invariants** — stable under cts. deformation

• **Rule of law** $\equiv$ **ergodicity** — $\bar{f}_t = \bar{f}_{\text{ens}}$

• **SQ** $\equiv$ **spectral gap** — $\lambda_1(\hat{L}_{ij})$

• **Crash** $\equiv$ **phase transition** — $T \rightarrow T_c^+$, $\xi \rightarrow \infty$

• $b \approx 1$ — Gutenberg-Richter from symmetry class

• **Somatic:** $\eta_{\text{som}}/\eta_{\text{cog}} \approx \kappa_{\text{EC}}^{-1}$

VI • Clinical Operators

ASC — high β , narrow Arnold tongue, deep attractors

CPTSD — non-ergodic, $\kappa_{\text{EC}} \ll 1$, EC decoupled

ADHD — high T , flat landscape, fast transitions

Co-occur: high $\beta \Rightarrow$ deeper trauma wells on adverse input

Tolerance window: $T_{\text{field}} \in [T_{\text{min}}, T_{\text{max}}]$

VII • Scale Map

σ 0-2 Quantum · 3-5 Cellular · 6-7 Organism · 8-9 Dyadic · 10 Geographic ($\tau_m \sim 10^{3-6}$ yr) · 11-14 Civilisational · 19 Cosmological (Λ)

VIII • Lean 4

✓ $\omega^2 = k/m$ · wave eq · FM-HN · $O(N^2)$ · WKB · T_c · BFSS $\times G_2$ holonomy · Zoom covariance (*honest axioms*)

ITI-Theory/U · doi.org/10.5281/zenodo.20460771